Hyperbolic Flavor Geometry: Conjectures on Dark Sector, Dark Energy, and Information Geometry

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Abstract

We present three conjectures extending the Hyperbolic Flavor Geometry (HFG) framework beyond the Standard Model flavor sector. The *Dark Sector Arithmetic Conjecture* predicts that dark matter particles correspond to non-Lucas primes appearing in covering towers of cusped hyperbolic 3-manifolds, with specific mass predictions (e.g., 1.85 MeV sterile neutrino, 39 MeV dark sector mediator, 580 MeV sub-GeV dark matter). The *Cosmological Unfolding Conjecture* identifies the transition from matter-dominated to dark-energy-dominated expansion with the spectral transition from slope-encoded (sparse) to universal geodesics in Dehn fillings of $m003$ and $m006$, and proposes a structural explanation for the Hubble tension. The *Ihara–Selberg Bridge* connects the geodesic length unit $\log \varphi$ to the information capacity of a binary tree, $k \log_2 \varphi$, suggesting a link between hyperbolic geometry and error-correcting codes. All conjectures are stated precisely enough to be falsifiable.

1 Introduction

The Hyperbolic Flavor Geometry (HFG) framework [1, 2, 3, 4] provides a geometric origin for the Standard Model flavor parameters: fermion masses ($m = m_e \varphi^{q/4}$), CKM and PMNS mixing matrices, and CP violation from $\mathbb{Z}/5$ torsion. The framework reduces the flavor sector to a single arithmetic object: the Lucas sequence $L_k = \varphi^k + \varphi^{-k}$ evaluated at quarter-integer indices $k \in \frac{1}{4}\mathbb{Z}$.

The present note extends the HFG framework to three speculative but falsifiable directions:

1. Dark sector particles identified by non-Lucas primes in covering towers.

2. Dark energy and the Hubble tension as manifestations of the spectral transition from sparse to universal geodesics.

3. An information-geometry correspondence linking the geodesic length unit $\log \varphi$ to the capacity of binary trees and error-correcting codes.

2 Dark Sector Arithmetic

Conjecture 1 (Dark Sector Arithmetic)**.** *Let* $\mathcal{P}_{\text{visible}} = \{2, 3, 5, 7, 11, 29, 47, 199, \ldots\}$ *be the primes appearing in the covering towers of the visible sector manifolds* M_{PMNS} *and* M_{CKM} *(prime Lucas numbers together with the ancestor torsion 5). Any prime* $p \notin \mathcal{P}_{\text{visible}}$ *that appears as torsion in the covering tower of a cusped hyperbolic 3-manifold is a candidate for a dark sector particle.*

Specifically, the primes $\{13, 17, 19, 23, 31, 41, 43, \ldots\}$ *observed in the control scan of* m004*,* m007*,* m009*,* m010*,* m015 *identify candidate dark sector manifolds. The mass of a dark fermion associated with such a prime satisfies:*

$$\frac{m_{\text{dark}}}{m_e} = \varphi^{q/4},$$

where $q/4$ *is a quarter-integer that is* not *a quarter-Lucas index (i.e., not in the set* $\{k/4 : L_k \in \mathbb{Z}\}$*). The covering tower prime* p *labels the manifold; the mass index* q *is a separate prediction requiring additional structure.*

2.1 Mass predictions for candidate primes

Using the mass lattice $m = m_e \cdot \varphi^{q/4}$, we list the masses corresponding to the smallest non-Lucas quarter-indices for each candidate prime, taking $q = p$ as a representative.

Table 1: Dark sector candidate masses from non-Lucas primes. All masses in MeV unless stated. $\varphi = (1 + \sqrt{5})/2$.

Prime p	$q = p$	$m \approx m_e \cdot \varphi^{p/4}$	Note
13	13	1.85 MeV	sterile neutrino range
17	17	4.8 MeV	dark photon range
19	19	6.4 MeV	light dark matter
31	31	39 MeV	dark sector mediator
41	41	580 MeV	sub-GeV dark matter
43	43	940 MeV	near proton mass

2.2 Falsifiability: the 17 MeV dark photon anomaly

The claimed 17 MeV dark photon anomaly (from ^{8}Be nuclear transitions) implies mass ratio $m/m_e \approx 33.3$. The corresponding quarter-index is

$$\frac{q}{4} = \frac{\log(33.3)}{\log \varphi} \approx 7.28, \quad q \approx 29.1.$$

Since $29 = L_7$ is a prime Lucas number, this anomaly falls in the *visible* sector in the HFG classification. The prediction is therefore that if the 17 MeV anomaly represents a genuine new particle, it is a visible-sector boson, not a dark sector particle.

3 Cosmological Unfolding and the Hubble Tension

Conjecture 2 (Cosmological Unfolding)**.** *The transition from matter-dominated to dark-energy-dominated cosmological expansion corresponds to the spectral transition from the sparse (slope-encoded) tier to the universal tier in the geodesic length spectra of Dehn fillings of* $m003$ *and* $m006$*.*

Let $n^*(m003) \approx 21$ *and* $n^*(m006) \approx 18$ *be the phase transition thresholds (Section 3 of the companion paper [5]). The matter-energy equality redshift* $z_{\text{eq}} \approx 0.3$ *corresponds to these thresholds, which in turn correspond to Lucas indices* $k^* \in \{6.0, 6.4\}$*, i.e.,* $L_{k^*} \approx 18$*–*23*. The dark energy fraction* $\Omega_\Lambda \approx 0.70$ *is numerically close to the average universal-tier occurrence at threshold:*

$$\bar{f}(n^*) = \tfrac{1}{2}(65.6\% + 81.8\%) \approx 73.7\%,$$

a suggestive but not yet derived coincidence.

3.1 Hubble tension direction

The lepton manifold M_{PMNS} enters the universal tier at $n \approx 21$; the quark manifold M_{CKM} enters at $n \approx 18$. If early-universe (CMB) measurements are dominated by quark-sector physics and late-universe (distance ladder) measurements by lepton-sector physics, there is a systematic offset in the inferred H_0.

The observed ratio $H_0^{\text{early}}/H_0^{\text{late}} \approx 67.4/73.0 \approx 0.923$ is in the correct direction: lepton-dominated measurements give lower H_0. The threshold ratio $n^*(m003)/n^*(m006) = 21/18 \approx 1.167$ (lepton threshold higher) is consistent with this direction.

Remark 3. *The ratio of squared thresholds* $(18/21)^2 \approx 0.735$ *is close to* $\Omega_\Lambda \approx 0.70$*. This numerical coincidence is recorded for future investigation without claiming a mechanism.*

3.2 Dehn filling flow model

A more ambitious formulation models cosmic expansion as a flow on Dehn filling parameter space: $(p(t), q(t)) \to (-2, 3)$ and $(-5, 2)$ as $t \to t_{\text{now}}$. Early universe ($|p| + |q|$ large): near-cusp, universal tier dominates, symmetric phase, no distinct particles. Current universe: specific slopes reached, sparse tier emerges, Standard Model particle content appears. The Hubble parameter is then

$$H(t) \propto \frac{d}{dt}\big[\log \operatorname{vol}(M(p(t), q(t)))\big],$$

decreasing as volume approaches the filled manifold values. Dark energy acceleration corresponds to the slowdown in volume change as the flow asymptotes to the target slopes.

4 Ihara–Selberg Bridge and Information Geometry

Conjecture 4 (Ihara–Selberg Bridge)**.** *The geodesic length spectrum of M_{PMNS} is the Riemannian analog of the Ihara zeta function of a rooted binary tree of depth $k_{\max} = 9$ (the covering tower degree). The information capacity of a binary tree of depth k is $C(k) = \log_2 F_k$, where F_k is the k-th Fibonacci number. As $k \to \infty$, $F_k \sim \varphi^k/\sqrt{5}$, so*

$$C(k) \sim k \log_2 \varphi.$$

The fundamental geodesic unit $\log \varphi$ and the information unit $\log_2 \varphi$ are the same object in different bases: $\log_2 \varphi = \log \varphi / \log 2$. The prime $2 = L_0$ appears at degree 3 in the covering tower of M_{PMNS}, linking binary information storage to the arithmetic topology of the manifold.

Remark 5. *Reed-Solomon error-correcting codes operate over finite fields $\mathrm{GF}(2^k)$. The base field $\mathrm{GF}(2)$ corresponds to the prime $2 = L_0$. The covering tower introduces this prime at degree 3, suggesting a connection between the degree-3 cover and 3-bit binary encoding. The covering tower of M_{PMNS} may be a geometric model for error-correcting codes with Fibonacci scaling, though the precise mechanism is not yet understood.*

5 Discussion

The three conjectures are speculative but precisely formulated. The most immediate falsifiable tests are:

- Search for dark matter at 1.85 MeV, 6.4 MeV, 39 MeV, 580 MeV, and 940 MeV; check whether mass ratios to m_e are Lucas or non-Lucas quarter-powers of φ.
- Verify that the 17 MeV anomaly (if real) is a visible-sector boson, not a dark sector particle.
- Model the Hubble tension using lepton/quark threshold ratio $n^*(m003)/n^*(m006) = 21/18$.
- Compute the information capacity of the M_{PMNS} covering tower and compare to known Fibonacci-based error-correcting code efficiencies.

Data Availability

All computations used SnapPy [6] version 3.3.2. Scripts: `https://github.com/drmlgentry/hyperbolic-flavor-scan`.

Conflict of Interest

The author declares no conflicts of interest.

References

[1] M. L. Gentry, *Quark Mixing from Hyperbolic Geometry*, Results in Physics, submitted (2026). SSRN: 10.2139/ssrn.6583550.

[2] M. L. Gentry, *Lepton Mixing from the Borel Structure of Hyperbolic Holonomy*, Results in Physics, submitted (2026). SSRN: 10.2139/ssrn.6583553.

[3] M. L. Gentry, *CP Violation from A-Factor Twist Angles of Hyperbolic Holonomy*, Results in Physics, submitted (2026). SSRN: 10.2139/ssrn.6583551.

[4] M. L. Gentry, *The Lucas Sequence as the Arithmetic Bridge in Hyperbolic Flavor Geometry*, J. Number Theory, submitted (2026). JNTH-D-26-00538. SSRN: 10.2139/ssrn.6660598.

[5] M. L. Gentry, *Spectral Universality and Slope-Encoded Arithmetic in Dehn Fillings of Hyperbolic 3-Manifolds*, in preparation (2026).

[6] M. Culler, N. M. Dunfield, M. Goerner, J. R. Weeks, *SnapPy*, `http://snappy.computop.org`.